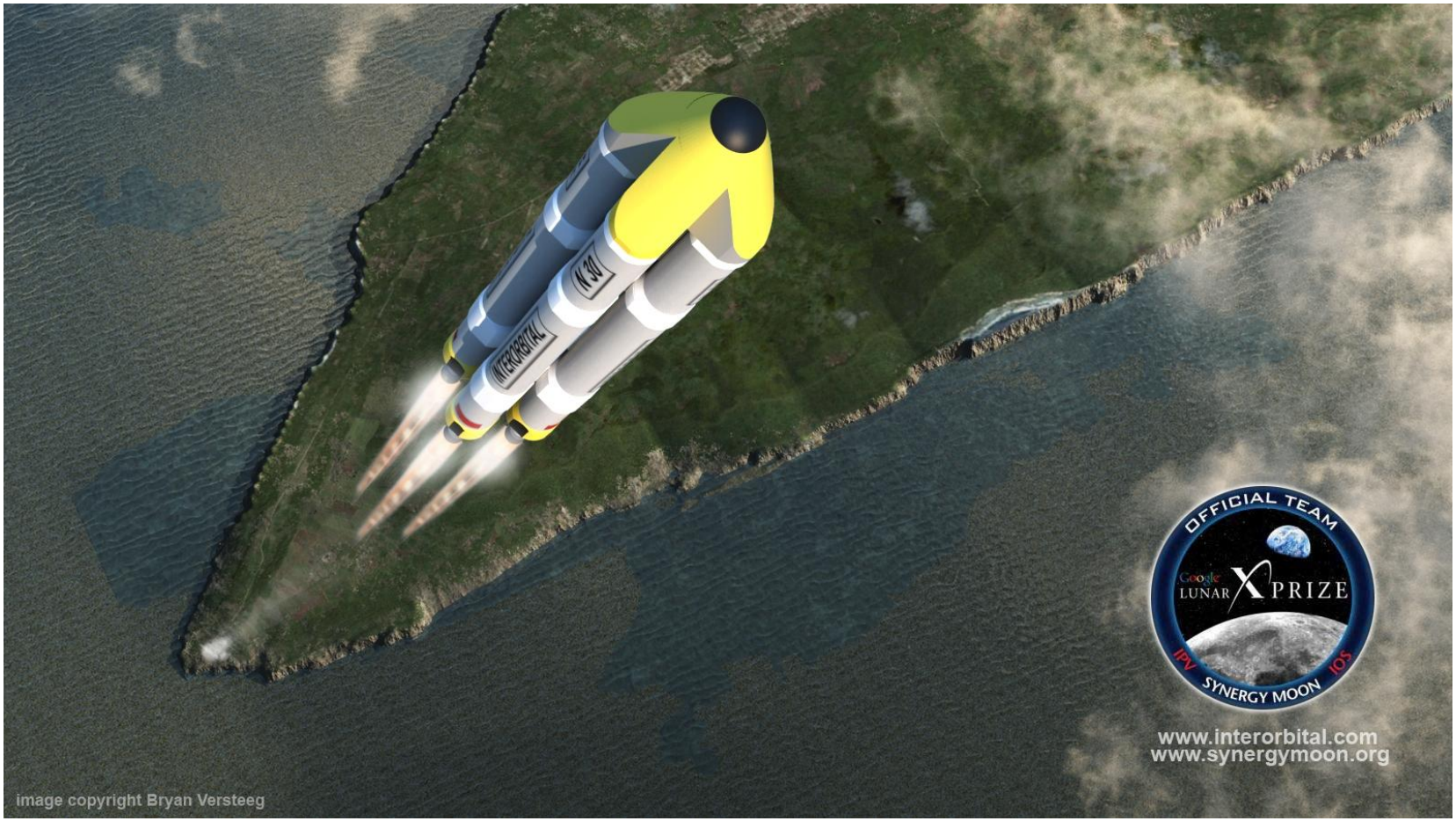
INTERORBITAL SYSTEMS' NEPTUNE MODULAR ROCKET SERIES:
Dedicated Small Sat Launchers / Low–Cost Orbital, Lunar, and Interplanetary Mission Enablers

Cal Tech/MIT Interplanetary Small Satellite Conference June 20-21, 2013 Pasadena, CA

Presenter: Randa Milliron, CEO/CoFounder, Interorbital Systems



Interorbital Systems

www.interorbital.com

CPM TEST VEHICLE on MOBILE ROCKET LAUNCHER

BREAKTHROUGH TECHNOLOGY

Storable Propellant Rocket Engine Technology

-- Successful test of attitude-control thruster and rocket controller

-- IOS is first in the US to use high-density nitric acid and turpentine as propellants of choice

--Substitutes for expensive, toxic hypergols like hydrazine and nitrogen tetroxide

GPRE 7.5KNTA CPM Main Engine Static Firing

-- Bi-propellant storable, high-density, hypergolic liquid rocket system: Nitric Acid / Turpentine

-- Thrust = 7,500 pounds at sea level

-- ISP 235 seconds (sea level); Density specific impulse 316 seconds (sea level); Vacuum ISP = 295 sec

-- Blowdown propellant feed; no ignition system or turbopumps required

-- State-of-the-art, all-composite combustion chamber and nozzle

-- Replaceable ablative chamber cartridge yields plug-and-play engine reusability

-- Designed for rapid mass production

COMMON PROPULSION MODULE (CPM)

-- Basic building block/construction element of the modular N-Series Rockets

-- Bi-propellant storable, hypergolic liquid rocket system

-- Blowdown propellant feed

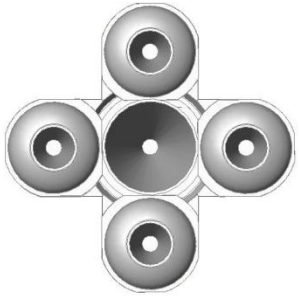
-- All-composite propellant tanks

-- Single gimbaled rocket engine; roll-control thrusters on single CPM units (3rd stage of N7 & N9 or 2nd stage of N5)

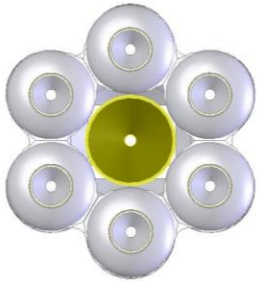
-- CPMs clustered together in multiples to meet mission requirements for both small and large payloads

-- Stand-alone sounding rocket SR145:145kg to 310km: Want to test your ion engine? Payload Space Available!

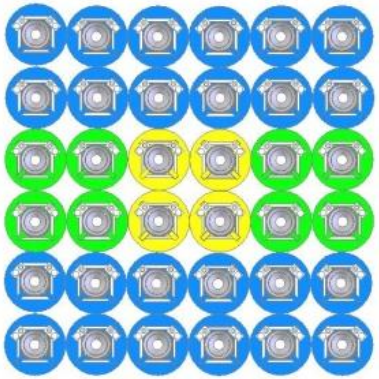
SELECT CPM CONFIGURATION EXAMPLES



N5
5 CPMs
Three stages
Payload: 30 kg



N7
7 CPMs
Three stages
Payload: 50 kg



N36
36 CPMs
Three stages
Payload: 1,000 kg

NEPTUNE 7 (N7) Dedicated Small-Sat Launcher

Three-Stage Orbital Launch Vehicle with Parallel Staging

-- Seven (7) Common Propulsion Modules (CPMs)

-- Stage 1: 4 CPMs: Thrust = 30,000 pounds

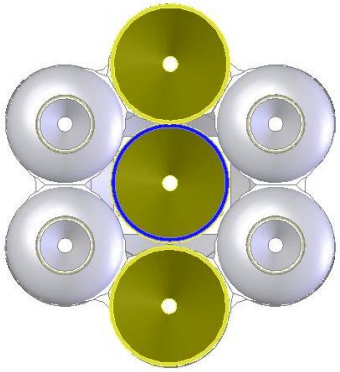
-- Stage 2: 2 CPMs: Thrust = 17,000 pounds

-- Stage 3: 1 CPM: Thrust = 8,500 pounds

-- Length: 36 feet (10.97 m); Maximum diameter: 6.2 feet (1.89 m)

-- Payload: 110.25 lbs (50 kg) to a 192 mile (310 km) polar orbit

-- Fits in 40-ft cargo container for easy transport



Advantages of Canister Ocean Launch

Ocean Launch Flexibility Advantage

-- Allows the customer to set the launch schedule

-- Safer for manned launches

-- Allows rocket to be positioned for any orbit

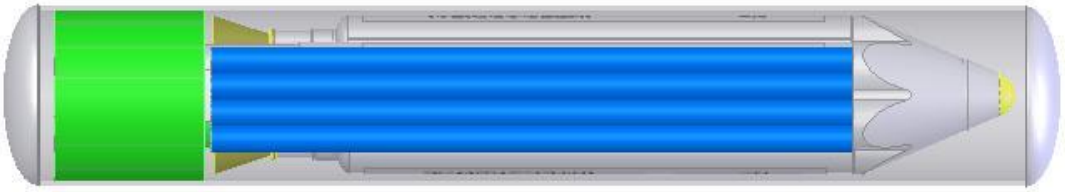
-- Doesn't set a limit on the size of a launch vehicle

-- Requires only a minimum of launch support hardware

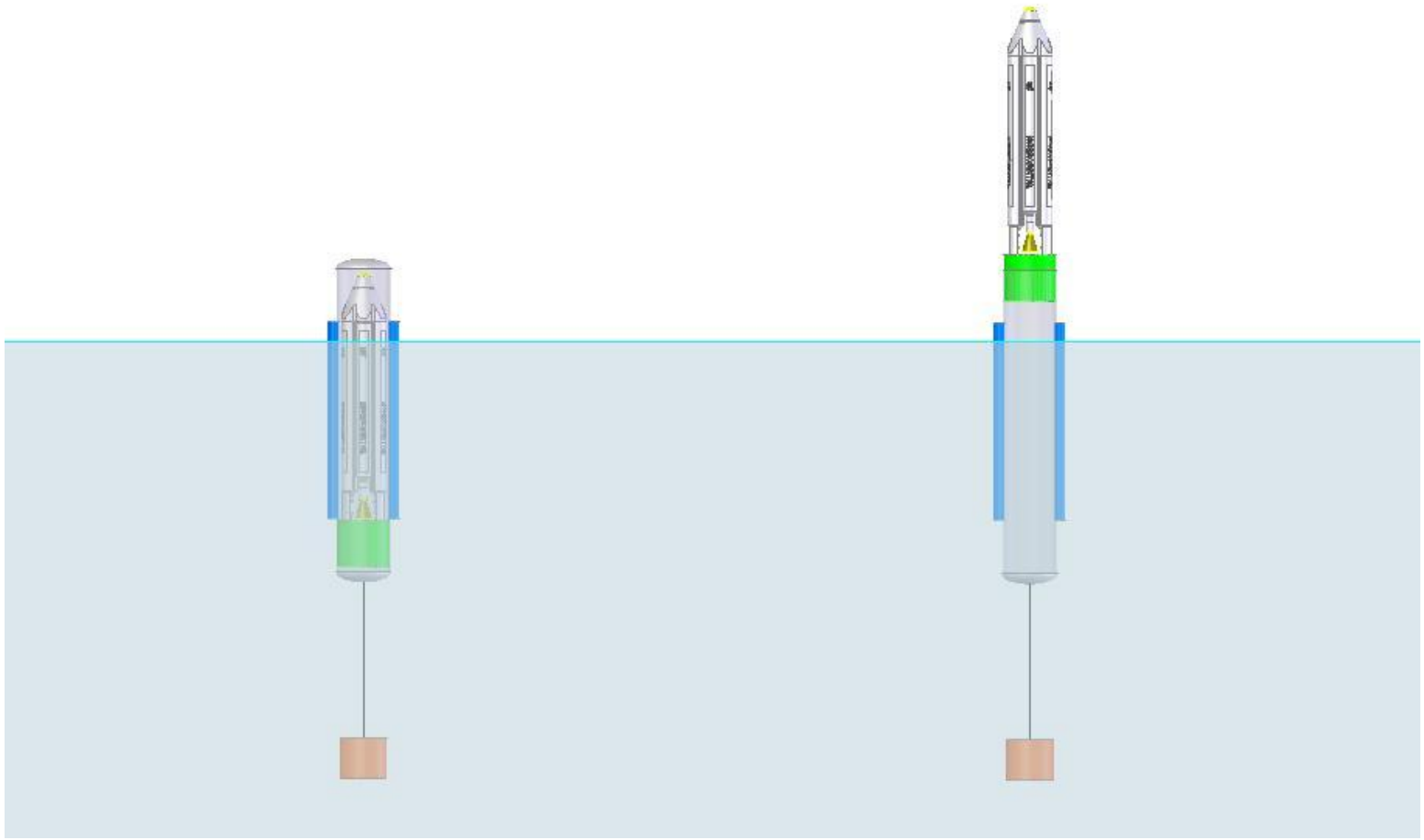
-- Rapid-response; no waiting in a spaceport line

-- The most cost-effective launch option

NEPTUNE Rocket Ocean-Based Canister Launch



NEPTUNE Canister-Launch Sequence



Ocean-Based Launch Site

-- Pacific Ocean off the coast of Southern California

-- 135 to 175 miles west of Los Angeles on the open ocean

-- Launch direction is to the south for both polar and sun-synchronous orbits

-- Launch site and orbit can be modified according to the customer's requirement

-- Dedicated to small sat polar and SSO launches

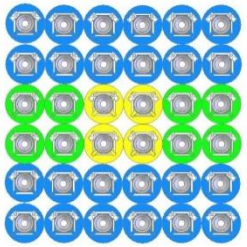
-- No Extortionate Spaceport Fees or Scheduling Problems

-- Makes Launch-on-Demand Possible!

Hawai'i Ocean Launch Option

Canister ocean-launch staged from the Big Island of Hawai'i is ideal for near-equatorial and lunar missions---and no ITAR issues! Interorbital's Google Lunar X PRIZE Team SYNERGY MOON launches are likely to originate from this location, which is also well-suited for interplanetary small-sat launches.

N36 Lunar Rocket



-- 36 Common Propulsion Modules (CPMs)

-- 36 Ablatively-Cooled Liquid Rocket Engines

-- 3-Stage Rocket + Payload: Satellite(s) or Lunar/ Interplanetary Transfer Vehicle)

-- Blowdown Pressure-Feed

-- Storable, High-Density, Hypergolic Propellants

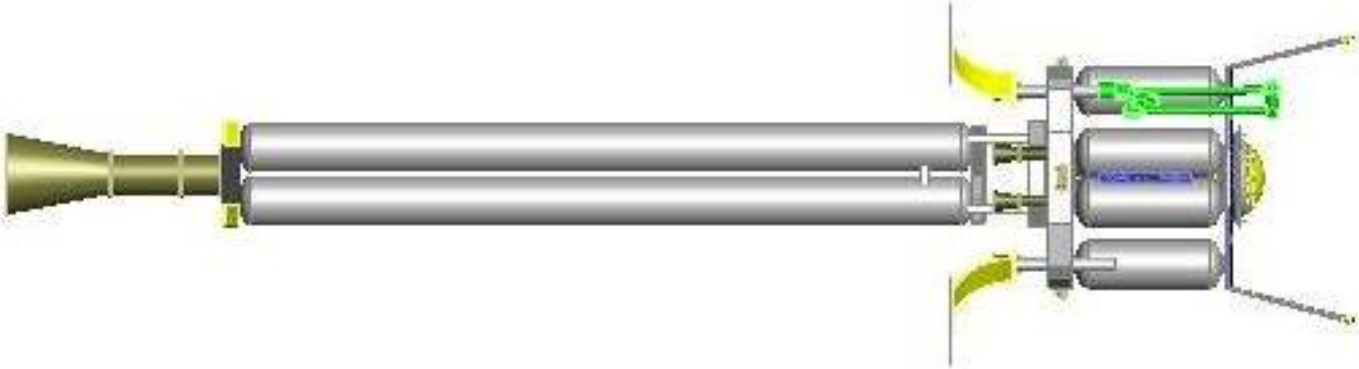
-- Modular Construction

-- Canister Ocean Launch

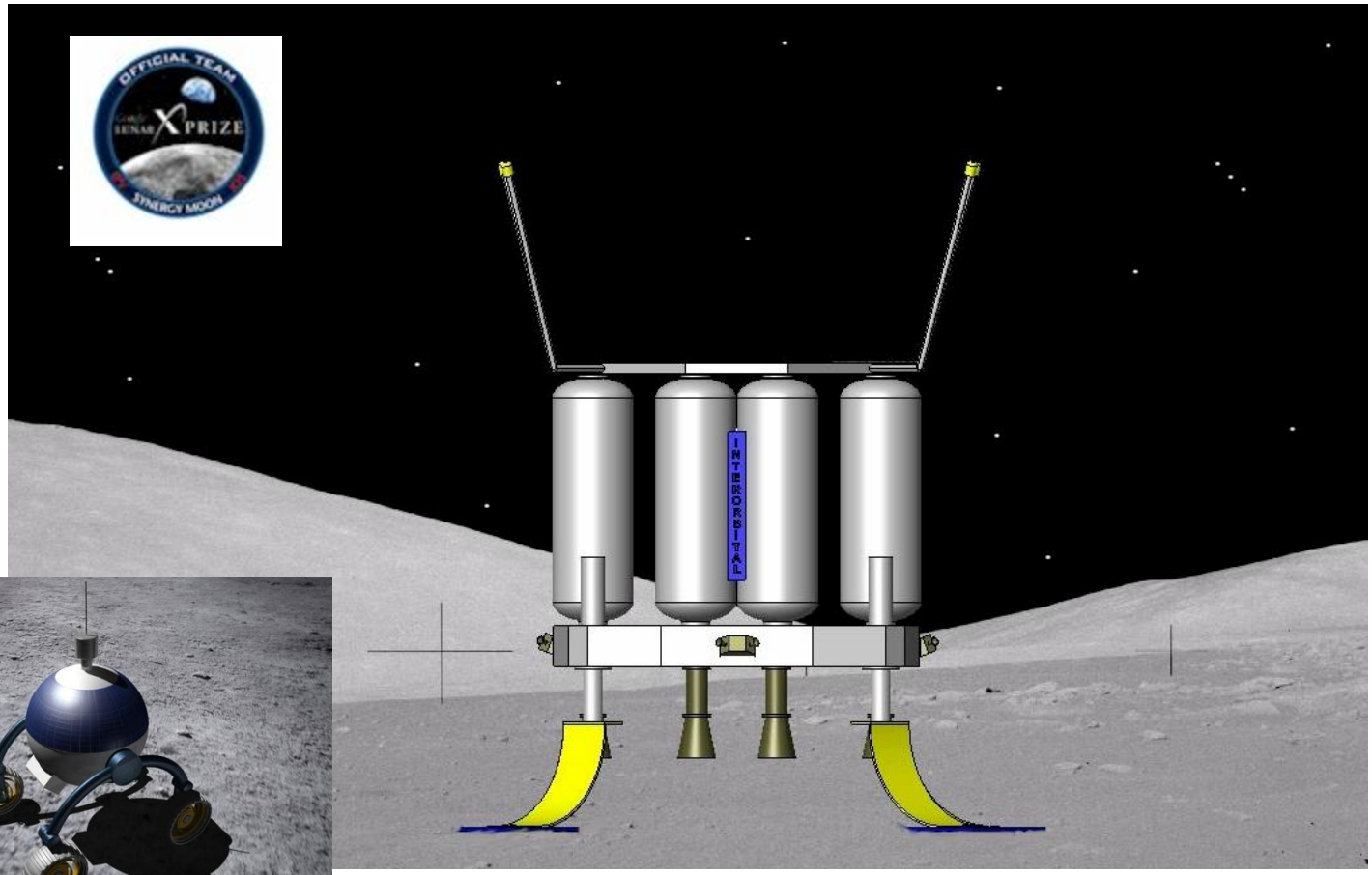
Maximum Thrust = 360,000 lbs

Payload to LEO: 1,000 kg (2,205 lbs) Lunar Surface: ~40kg

Lunar Transfer Stage



GLXP Team SYNERGY MOON Lunar Lander and Rover



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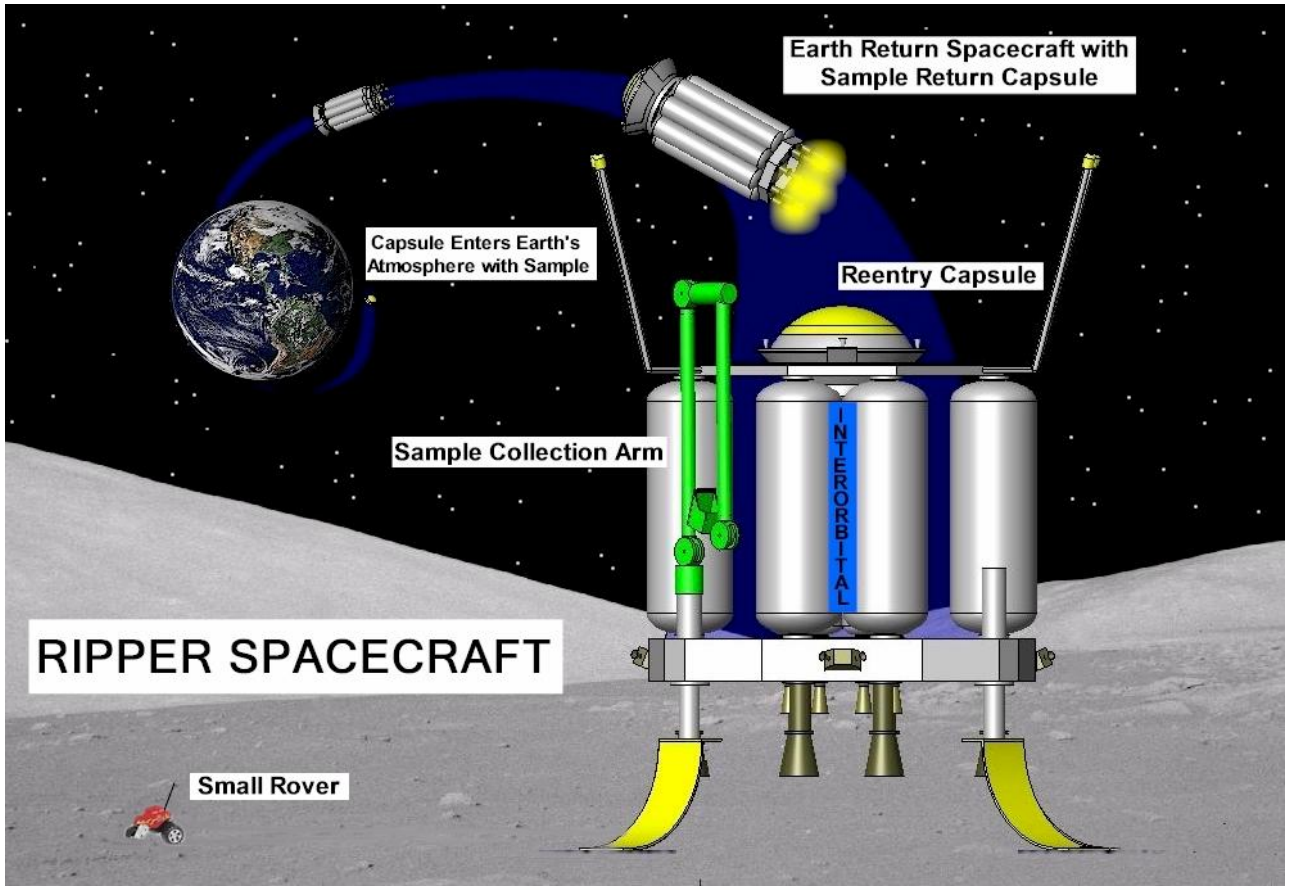
IOS Lunar Sample Return Mission



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INTERORBITAL'S UPCOMING MISSIONS

Completed Phase I NASA SBIR Small Business Innovation Research Award 2012

CPM TV: Low-altitude suborbital test flights under FAA Class 3 Waiver 2013
SR 145: High-altitude sounding rocket launch: 145kg to 310km 2013

Olav Zipser Rocket Diver High-Altitude Jump Record Attempt from SR 145 CPM

NEPTUNE Small-Sat Orbital Missions I and II 2014
Google Lunar X PRIZE Lunar Missions 2015
Private-Sector Lunar Sample/Return Mission 2015-16
Orbital Expeditions Space Tourism Flights 2015-16

IOS TUBESAT/CUBESAT PERSONAL SATELLITE KITS

-- PCB Gerber Files

-- Spectrolab TASC solar cells

-- A Li-ion battery pack (3.7 V 5200 mAh)

-- Microcomputer (NetMedia BasicX-24 or Arduino Mini)

-- Transceiver (Radiometrix)

-- Antennas,

-- Fasteners

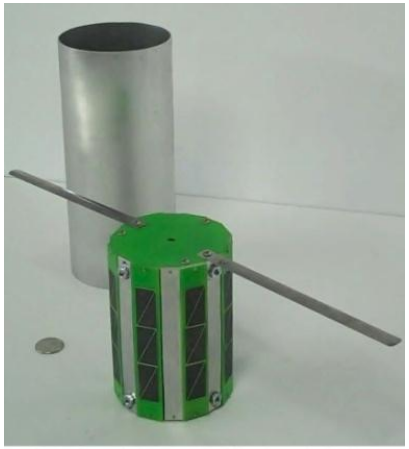
-- Complete instructions and assembly guide

-- Academic base-price for IOS TubeSat kit is $8,000

-- Academic base-price for IOS CubeSat kit is $19,125

-- 24 Units or single satellite equivalent per launch

IOS CubeSat Kit



TubeSat with Sample Ejection Cylinder

All kit prices include a launch to orbit on a NEPTUNE rocket!

CubeSats

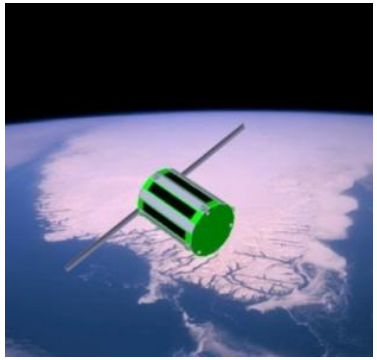
UC Irvine, UCISAT1
FPT University, Vietnam, F-1 CubeSat
Nanyang Technological University, Singapore VELOX-P CubeSat,
Google Lunar X PRIZE(GLXP) Team PLAN B (Canada)
GLXP Team EuroLuna, Romit 1 (2-Unit CubeSat from Denmark)
New! GLXP Team SYNERGY MOON: Tesla Telescope Project (3U)
NASA Independent Verification and Validation (IV&V) Facility, 1 CubeSat & 2 TubeSats
King Abdullah University, Saudi Arabia (KAUST) (2 IOS CubeSats;1TubeSat; 1 suborbital
The Golden iPod: Voyager revisited! Earth-to-Sky/spaceweather.com; Bishop, CA
New! Pakistan's Islamabad Institute of Science and Technology
Taiwan's National Cheng Kung University, 2U TARO-2

Denmark's GLXP Team Euroluna: Romit 1, 2U CubeSat

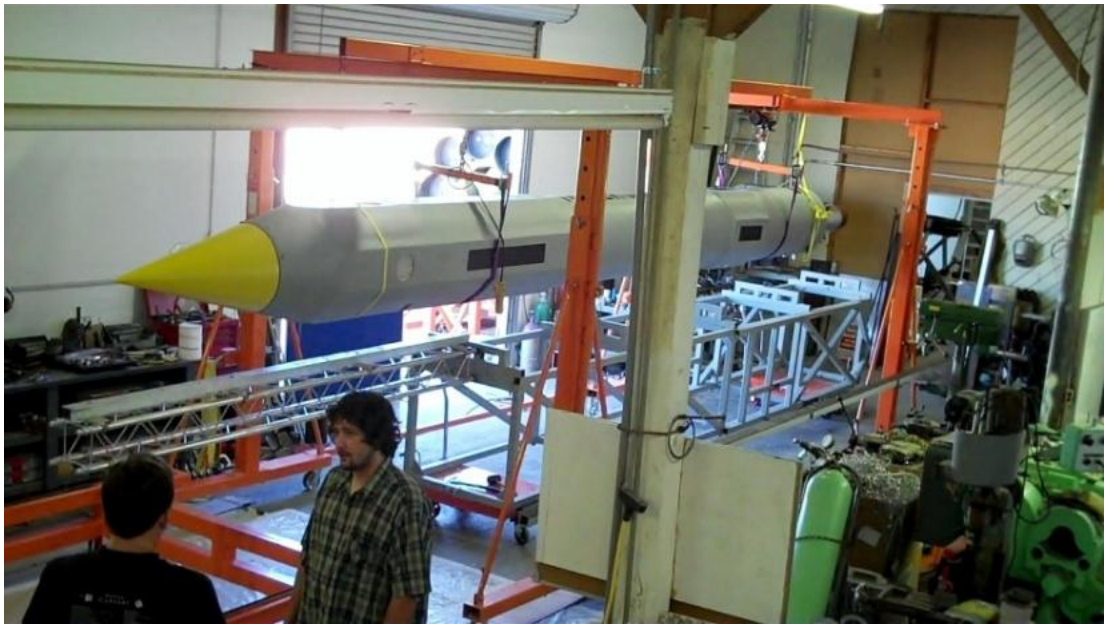
NEPTUNE: MISSION I & II LAUNCH MANIFESTS

TubeSats

Morehead State University (Kentucky Space) (TubeSat and 2 suborbital payloads)
InterAmerican University of Puerto Rico; STEM Program
University of Sydney (Australia) (2) *i-INSPIRE (initial-INtegrated SPectrograph, Imager & Radiation Explorer)*
Aslan Academy (Private LA High School) STEM Program
Project Calliope (Space Music Project) Dr. Sandy Antunes' Mission to Sonify the Ionosphere
Universidad de Puerto Rico / Marcelino Canino Canino Middle School, STEM micro-meteoroid impact study
GLXP Team SYNERGY MOON Space-Qualifying Rover Team Astronomska Udruga Vidulini's (AUV) Comms
GLXP Team Part-Time Scientists / Fluid & Reason Software (2) (US/Germany): Wes Faler's FRETS 1
Naval Postgraduate School (3) (TubeSats as ad-hoc orbital communication nodes) and 2 suborbital payloads
Defense Science and Technology Lab (DSTL) United Kingdom; Earth Observation/Remote Sensing
Austrian Arts Group mur.at with MURSAT-1: Earth-as-Art Project
United States Military Academy at West Point (2) STEM Program
Brazilian Space Institute/108 5th-7th Grade Students, Ubatuba, Sao Paulo, Brazil STEM Program
Mexican Satellite Project ULISES-1 Sat from PLAY Festival's Arts/Soccer Opera from Space
TriVector Services (Huntsville) TRACsat – TriVector Radiation and Attitude Control Satellite
Diverbo.es/Iniciativas en Idiomas (Madrid, Spain)
NASA Independent Verification and Validation (IV&V) Facility (2); STEM Program
Galaxy Global, 1 TubeSat, donated to NASA Educational Program
Institute of Advanced Media Arts & Sciences (IAMAS)/The Science Project, Inc. Japan (7)
AKQA Advertising, San Francisco
Universidad de Chile, Santiago
New! University of Sao Paulo, Laboratory of Integrated Systems, Brazil (2)
New! David Lawrence K-8 School, North Miami, Florida; STEM Program Curriculum
New! RADG--- Undisclosed Advertising Project
New! OMNI LABS (Brazil): 'Project Transcendence' Celebration of Art, Science, and Humanity
New! 4-H/Ute Mountain Youth/Colorado State University Extension STEM Satellite Project
New! KEN KATO, Private Satellite Project, Japan



Autumn 2013 Suborbital Launch Opportunity: 145kg to 310km Ocean-Launched Full-Performance SR145 CPM



INTERORBITAL'S ROCKET TECHNOLOGY TESTING, FIELD TRIALS, and LAUNCH CREW TRAINING